Deep Learning in Practice

Practical Session 3: Graph Attention Neural Networks

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Architecture

We use a Graph Attention Network (GAT) architecture based on *"Graph Attention Networks"* by Veličković et al. It consists of 2 self-attention layers with 4 heads and hidden size 256 (concatenating its outputs), and ELU activation between them followed by another self-attention layer with 6 heads with 121 outputs each. These outputs are averaged and followed by a logistic sigmoid activation. Skip connections are used in the intermediate layer.

We first show a detailed diagram of the attention mechanism and then the full model made with this building block.

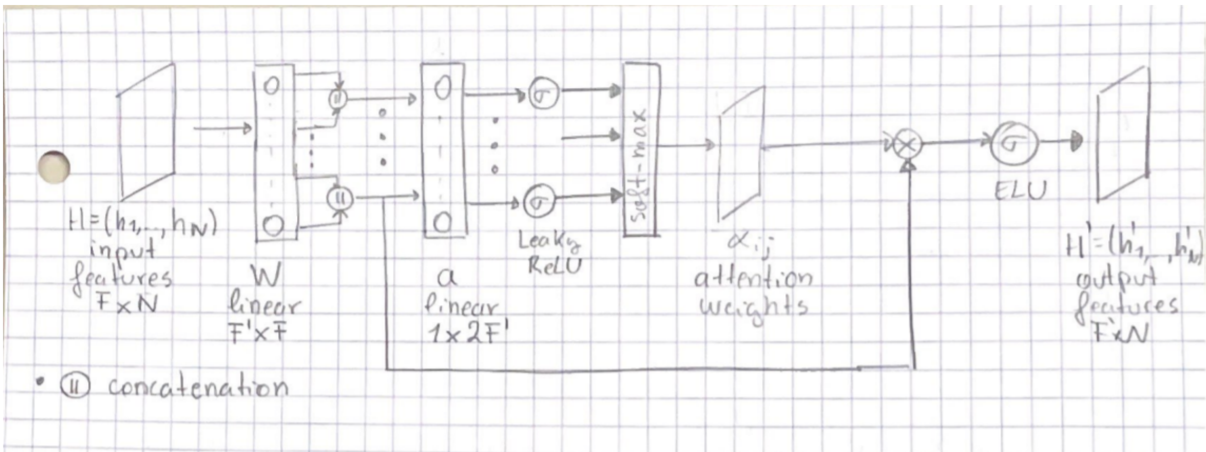


Figure 1: Attention mechanism

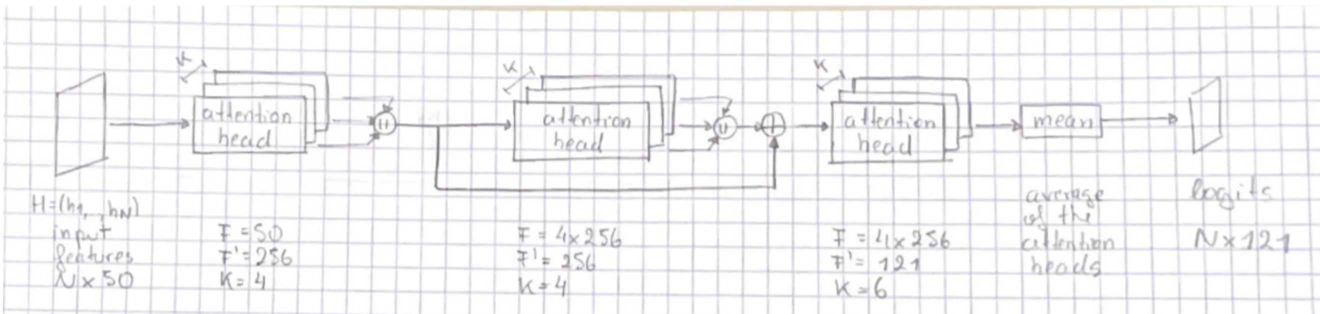
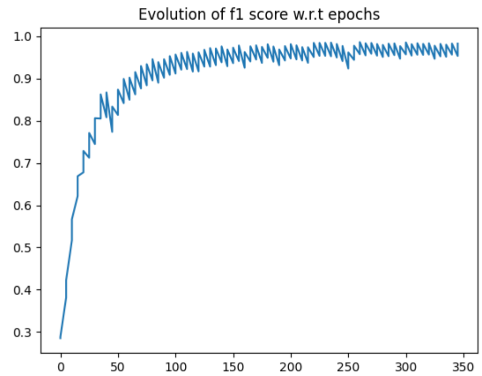


Figure 2: Complete architecture

We train the model for 350 epochs with early stopping on the F1-score of the validation dataset, actually stopping training on epoch 261. We use Adam optimizer with a learning rate of 0.005 and batch size 2.

Results

The F1-score is the harmonic mean between precision and recall. It summarizes into a single number the trade-off between precision and recall. In this multiclass problem, we report the micro average: a global average F1-score, weighted by the class frequency.



We obtain a score of **0.9700** on the test dataset, clearly outperforming the baseline of 0.5053, and close to the possible maximum score of 1.000.

Why is GAT better?

GAT has more expressive power since it uses different weights for different nodes (learned with the self-attention mechanism) which allow nodes to attend over their neighborhood depending on the nodes' features. In contrast, the graph convolutional model uses fixed weights based on the number of neighborhoods of the nodes.